# AI ASSISTED CODING
# Assignment-21.2

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Task 1 – AI-Assisted Pair Programming

• Use AI to generate an initial program (e.g., a simple To-Do List Manager).

• Ask AI to suggest enhancements (such as error handling, additional features, or refactoring).

• Compare the original and improved versions to understand the role of AI in collaboration.

Code screenshot:

```python
class ToDoList:
    def __init__(self):
        self.tasks = []

    def add_task(self, task):
        self.tasks.append(task)
        print(f'Task "{task}" added.')

    def remove_task(self, task):
        if task in self.tasks:
            self.tasks.remove(task)
            print(f'Task "{task}" removed.')
        else:
            print(f'Task "{task}" not found.')

    def view_tasks(self):
        if not self.tasks:
            print("No tasks in the list.")
        else:
            print("To-Do List:")
            for idx, task in enumerate(self.tasks, 1):
                print(f"{idx}. {task}")
```

Enhanced code screenshot:

```python
# Enhanced Program with AI Suggestions
class ToDoList:
    def __init__(self):
        self.tasks = []

    def add_task(self, task):
        if task in self.tasks:
            print(f'Task "{task}" already exists.')
        else:
            self.tasks.append(task)
            print(f'Task "{task}" added.')

    def remove_task(self, task):
        if task in self.tasks:
            self.tasks.remove(task)
            print(f'Task "{task}" removed.')
        else:
            print(f'Task "{task}" not found.')

    def display_tasks(self):
        if not self.tasks:
            print("No tasks in the list.")
        else:
            print("To-Do List:")
            for idx, task in enumerate(self.tasks, start=1):
                print(f"{idx}. {task}")

    def clear_tasks(self):
        self.tasks.clear()
        print("All tasks cleared.")
```

Task 2 – Version Control Integration

Initialize a Git repository for your project. Perform the following steps:

• Create and switch to a new branch.

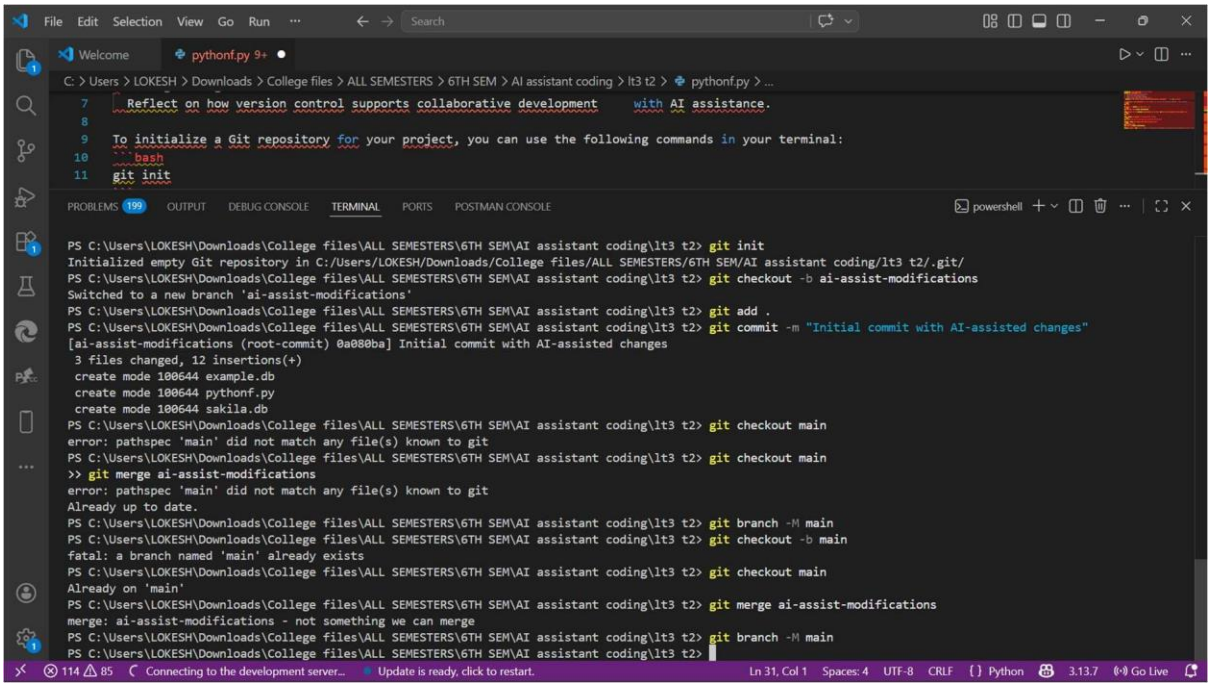
• Add AI-assisted modifications.

• Commit with a meaningful message.

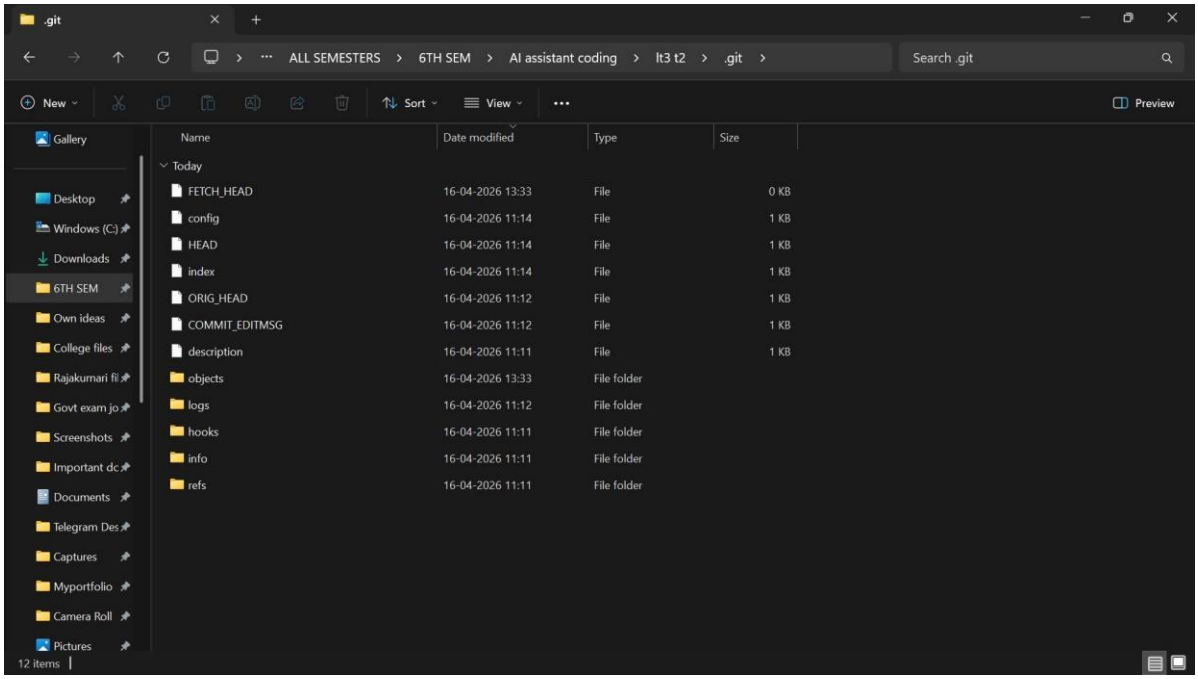
• Merge changes back into the main branch.

Reflect on how version control supports collaborative development with AI assistance.

Code screenshot:




Output:




**Task 3 – AI for Commit Message Generation**

- Use AI to generate **concise, meaningful commit messages** based on the changes made.
- Compare AI-generated messages with human-written ones.
- Discuss how AI can improve **documentation and traceability** in Git history.

Code screenshot:

```python
import os
import subprocess
def get_git_diff():
    """Get the git diff of the current repository."""
    try:
        result = subprocess.run(['git', 'diff', '--cached'], stdout=subprocess.PIPE, stderr=subprocess.PIPE, text=True)
        if result.returncode != 0:
            print("Error getting git diff:", result.stderr)
            return None
        return result.stdout
    except Exception as e:
        print("Exception occurred while getting git diff:", str(e))
        return None
def generate_commit_message(diff):
    """Generate a commit message based on the git diff."""
    if not diff:
        return "No changes to commit."

    # Simple heuristic for generating commit messages
    lines = diff.splitlines()
    added_lines = [line for line in lines if line.startswith('+') and not line.startswith('+++')]
    removed_lines = [line for line in lines if line.startswith('-') and not line.startswith('---')]

    if added_lines and removed_lines:
        return "Updated code with additions and deletions."
    elif added_lines:
        return "Added new code."
    elif removed_lines:
        return "Removed existing code."
    else:
        return "Modified code without clear additions or deletions."
def main():
    diff = get_git_diff()
    if diff is not None:
        commit_message = generate_commit_message(diff)
        print("AI-generated commit message:", commit_message)
if __name__ == "__main__":
    main()
```

Output:

```
ALL SEMESTERS/6TH SEM/AI assistant coding/22.5/Aiac_task2.py"
AI-generated commit message: No changes to commit.
```

**Task 4 – Pair Programming Simulation**

- Student A writes the initial version of a program.
- Student B, with AI assistance, suggests improvements or adds features.

Both students collaborate using Git branching and merging to integrate their work.

Code screenshot:

```python
# Student A's initial version of the program
# A simple program to calculate the area of a rectangle
def calculate_area(length, width):
    return length * width
# Example usage
length = 5
width = 3
area = calculate_area(length, width)
print(f"The area of the rectangle is: {area}")
# Student B's improvements and feature additions
# Student B suggests adding input validation and a feature to calculate the perimeter
def calculate_area(length, width):
    if length <= 0 or width <= 0:
        raise ValueError("Length and width must be positive numbers.")
    return length * width
def calculate_perimeter(length, width):
    if length <= 0 or width <= 0:
        raise ValueError("Length and width must be positive numbers.")
    return 2 * (length + width)
# Example usage with improvements
length = 5
width = 3

try:
    area = calculate_area(length, width)
    perimeter = calculate_perimeter(length, width)
    print(f"The area of the rectangle is: {area}")
    print(f"The perimeter of the rectangle is: {perimeter}")
except ValueError as e:
    print(e)
# Git branching and merging
# Student A creates a branch for the initial version
# git checkout -b initial-version
# Student A commits the initial version
# git add Aiac_task2.py
# git commit -m "Initial version of the rectangle area calculator"
# Student B creates a branch for improvements
# git checkout -b improvements
# Student B commits the improvements
# git add Aiac_task2.py
# git commit -m "Added input validation and perimeter calculation"
# Student A merges the improvements into the main branch
# git checkout main
# git merge improvements
# The final version of the program now includes both the area and perimeter calculations with input validation.
```

Git repository commands:

```
untracked files present (use "git add" to track)
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout main
ot match any file(s) known to git
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout -b initial version
mmit and a branch 'initial' cannot be created from it
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git init
tory in C:/Users/LOKESH/Downloads/College files/ALL SEMESTERS/6TH SEM/AI assistant coding/22.5/.git/
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git add Aiac_task2.py
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git commit -m "Initial commit - rectangle area calculator"
c] Initial commit - rectangle area calculator
ns(+)
k2.py
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git branch -M main
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout -b initial-version
itial-version'
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout main

\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout -b improvements
provements'
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git add Aiac_task2.py
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git commit -m "Added validation and perimeter calculation"


o include in what will be committed)


untracked files present (use "git add" to track)
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git commit -m "Added validation and perimeter calculation"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untracked files present (use "git add" to track)
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git commit -m "Added validation and perimeter calculation"

ree clean
\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git checkout main

\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5> git merge improvements

\College files\ALL SEMESTERS\6TH SEM\AI assistant coding\22.5>
```